TrueJES: the jet energy scale along the truth ladder

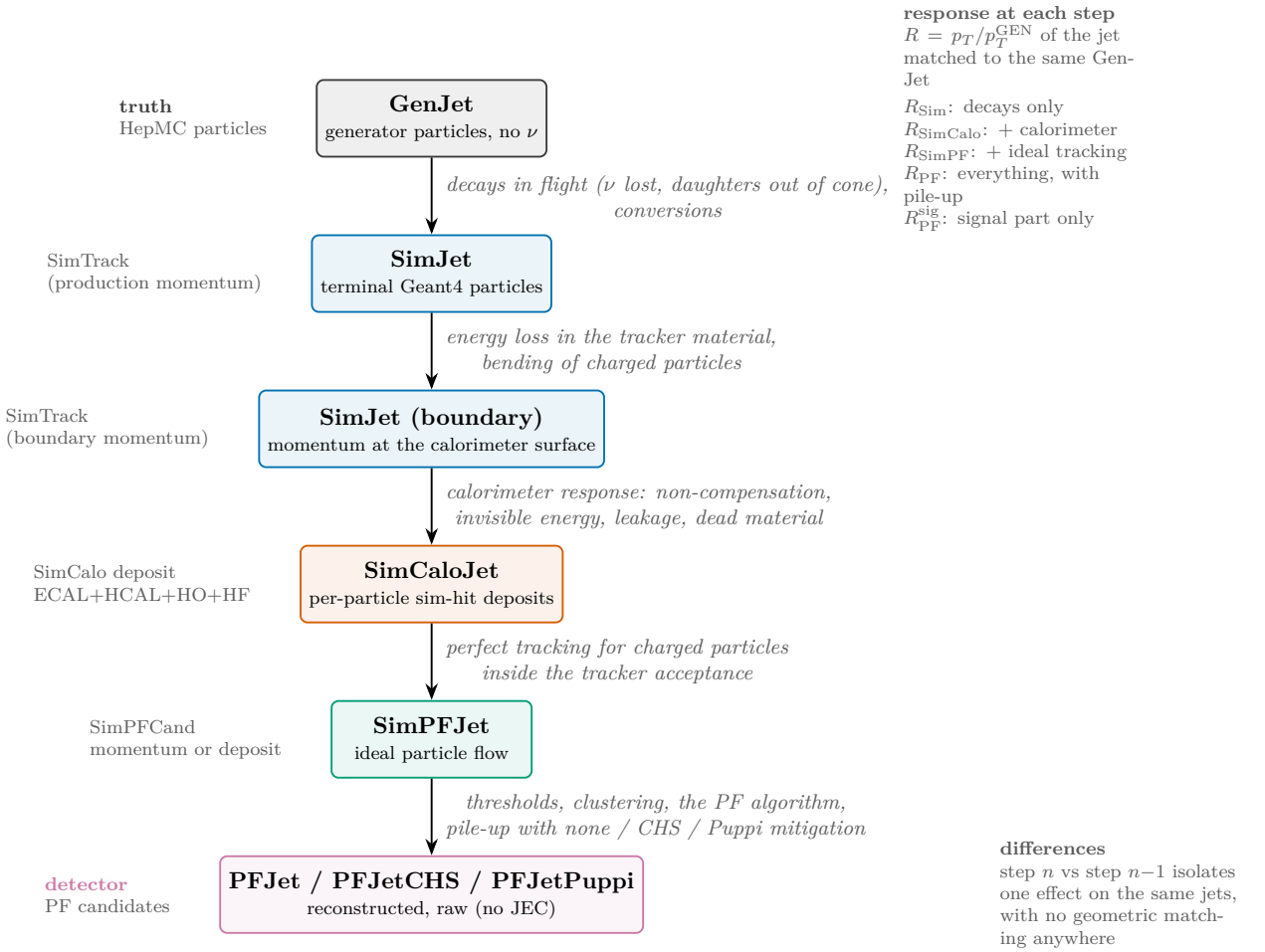
illustrations

Reco2Pico / analysis/TrueJES

October 16, 2026

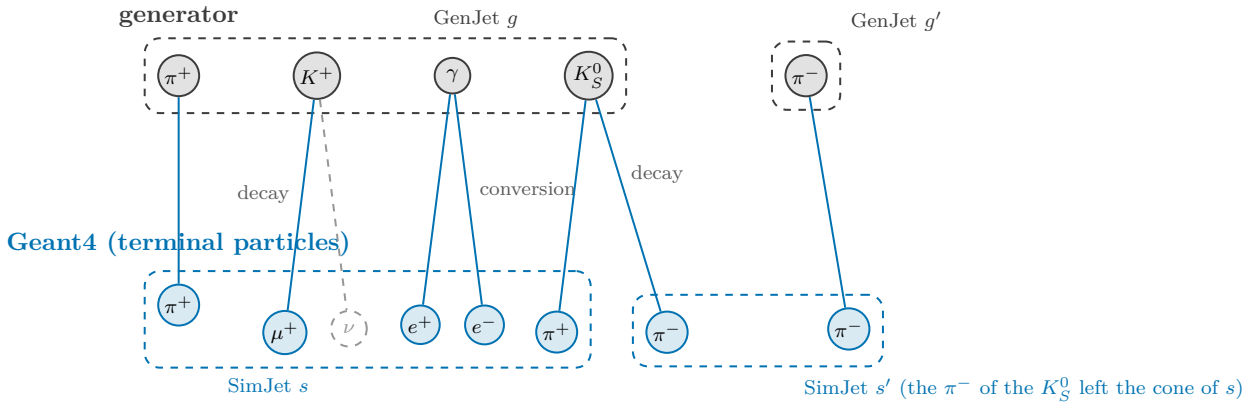
1. The ladder

One jet collection per step, all built on the same simulated events and all matched to the same generator jet. The difference of the response between two consecutive steps isolates one effect.



2. Matching by ancestry, not by ΔR

Every Geant4 particle descends from one generator primary, and every generator particle sits in at most one gen jet. The energy a jet shares with a gen jet is therefore exact, whatever the magnetic field and the decays did to the directions. The same scores are used at every step.



every SimTrack has one generator ancestor (walk `SimTrack_parentIdx` up to the primary) **every generator particle sits in at most one GenJet** (gen jets re-clustered in the job)

for a pair (s, g): $E_{\text{shared}} = \sum_{c \in s,\ \text{anc}(c) \in g} E_c = E_{\pi^+} + E_{\mu^+} + E_{e^+} + E_{e^-} + E_{\pi^+}$

`simFraction` $= E_{\text{shared}}/E(s)$: purity of s `genFraction` $= \dfrac{1}{E(g)} \sum_{p \in g} E_p\, f_s(p)$: efficiency of g into s, with $f_s(K^0_S) = \dfrac{E_{\pi^+}}{E_{\pi^+} + E_{\pi^-}}$, $f_s(K^+) = 1$ (the ν is not clustered)

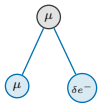
matched: the best pair for g with both fractions > 0.5; `ptRatio` $= p_T(s)/p_T(g)$ is the response of the pair

no ΔR anywhere: immune to the bending in the field and to the decay kinematics at low p_T

3. Which Geant4 particles are clustered

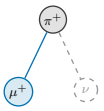
The terminal particles of the stored decay tree, with an energy clause so that a continuing particle that radiated a soft secondary is not removed. The same rule defines the ideal particle-flow list; Geant4's boundary-crossing flag is not used because particles born inside the calorimeter never carry it.

A. continuing particle



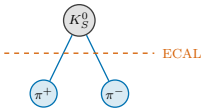
stored daughters carry $< 50\%$ of the energy (here 0.01%: ionisation in the muon system)
$\Rightarrow$ the μ **is terminal**, clustered with its full momentum; the δe^- is a constituent too

B. decay / conversion



daughters carry 100% of the energy
$\Rightarrow$ the π^+ is **replaced** by the μ^+; the ν is dropped (as in GenJetNoNu)
this is where the decay-in-flight response comes from

C. born inside the calorimeter

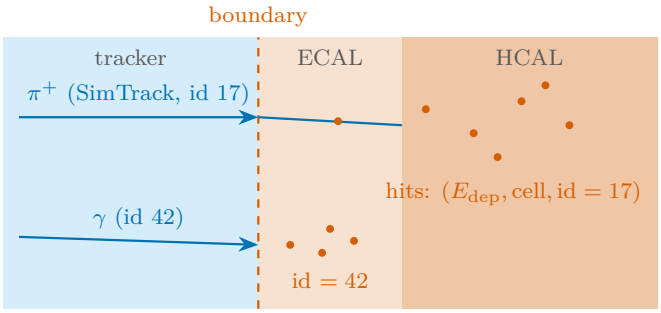


the pions never cross the tracker/calo boundary, so Geant4's `crossedBoundary` flag is not set, but the sim hits reference them
$\Rightarrow$ the terminal rule, not the flag, defines the particle lists (jets, ideal PF)

terminal $\Leftrightarrow$ no stored daughter, or $\sum_{\text{daughters}} E_d < 0.5\,E$ (`SimTrack_dauEnergyFrac` < 0.5)

4. The calorimeter response from the sim hits

The calorimeter sim hits carry the id of the particle that entered the calorimeter, so the deposits are summed per particle without clustering; jets of the deposits give the calorimeter-only response, and an ideal particle flow built from momenta and deposits shows what tracking recovers.



SimCalo: one row per particle that entered `geantTrackId` of every PCaloHit = the particle at the calo surface (CaloSD)
⇒ **no clustering**: sum the hits per id

ECAL hits in GeV; HCAL hits relabelled and × digitiser sampling factors (GeV-equivalent); HF photoelectrons × HF factor (unvalidated)

cells summed over all particles first → optional per-cell threshold (rechit-like) → shared in proportion to the deposits

`response` = $E_{\text{deposit}}/E_{\text{at boundary}}$ (single particle)
`SimCaloJet` = anti-k_T 0.4 of the deposits (direction = energy-weighted cell direction)

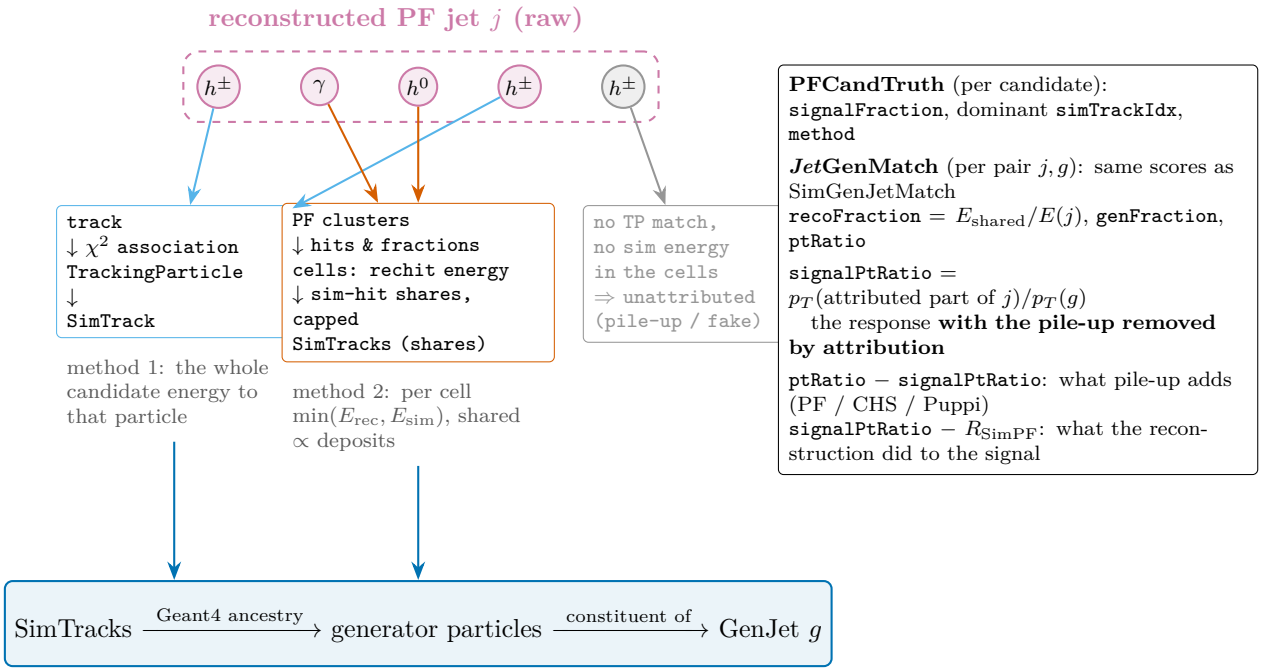
SimPFCand: ideal particle flow

terminal particle, charged, produced inside the tracker volume, $|\eta| < 2.5$, $p_T > 0.5$ GeV
→ its **momentum** (perfect tracking), kind 1
otherwise → its **deposit** (neutral: kind 2, charged not trackable: kind 3), or nothing if it left none

`SimPFJet` = anti-k_T 0.4 of these; SimPF − SimCalo = what tracking recovers

5. The reconstructed jets on the same ladder

PF candidates are attributed to simulated particles (track association for charged ones, sim-hit shares of the cluster cells for the others); what is not attributed is pile-up, noise or fakes. The reconstructed jets are then matched by the same ancestry scores, and the signal-only response separates the pile-up from the reconstruction effects without matching pile-up conditions.



PFCandTruth (per candidate): `signalFraction`, dominant `simTrackIdx`, `method`

***Jet*GenMatch** (per pair j, g): same scores as SimGenJetMatch
`recoFraction` = $E_{\text{shared}}/E(j)$, `genFraction`, `ptRatio`

`signalPtRatio` = p_T(attributed part of j)/$p_T(g)$
the response **with the pile-up removed by attribution**

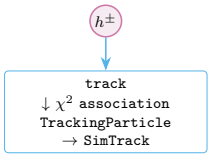
`ptRatio` − `signalPtRatio`: what pile-up adds (PF / CHS / Puppi)
`signalPtRatio` − R_{SimPF}: what the reconstruction did to the signal

6. How the pile-up is removed: attribution, not subtraction

Every PF candidate is split into the part that can be traced to a simulated particle of the hard-scatter event and a remainder. Charged candidates are traced through their track; neutral ones cell by cell

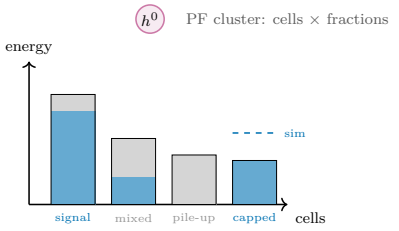
through the simulated deposits, capped by the simulated energy of the cell. The remainder is pile-up, noise or a fake. The signal part of a jet is the sum of the traced parts of its constituents, so no pile-up matching between samples is needed.

charged candidate $h^{\pm}$



match ⇒ **100%** of the candidate energy is signal
no match ⇒ **0%** (pile-up track, fake)
TrackingParticles exist for the hard-scatter event only

neutral candidate γ / h^0



sum over the cells, rescaled to the candidate energy
⇒ `signalFraction` of the candidate
the sim hits exist for the hard-scatter event only

the jet

$$\vec{p}_T^{\,\mathrm{sig}} = \sum_{c \,\in\, \mathrm{jet}} w_c \, f_c^{\mathrm{sig}} \, \vec{p}_{T,c}$$

w_c: clustering weight (Puppi), f_c^{sig}: `signalFraction`

`signalPtRatio` = $|\vec{p}_T^{\,\mathrm{sig}}|/p_T^{\mathrm{GEN}}$
`ptRatio` − `signalPtRatio` = the pile-up in this jet

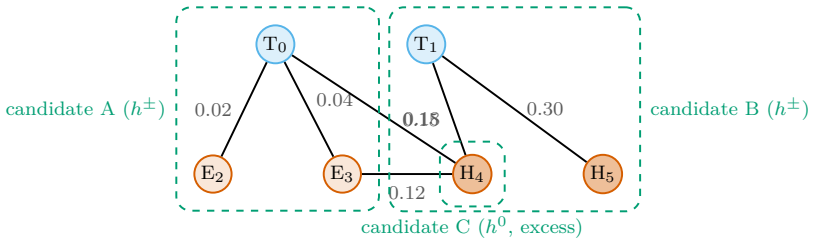
shared-energy response: keep only the shares whose particle descends from the matched GenJet's constituents
⇒ the gen jet's own energy, nothing else

signal = any particle of the hard-scatter event (underlying event and other jets included); nothing is subtracted, pile-up is simply never counted

7. The particle-flow block tables

For the reconstruction studies the Pico also keeps the full PF block structure: the elements, the geometric links PF found between them, and the elements each candidate finally used.

one PF block: elements and stored links



`PFBlockElem` – the inventory
one row per element: type, block, pointers to Track / PFCluster rows, candIdx

`PFBlockLink` – what PF found connected
one row per stored link: the two elements and the linking distance ($\Delta\eta$-$\Delta\phi$ at the calo entrance, placeholders for same-object links)
input of the algorithm, not its decision

`PFCandElem` – what PF decided
one row per (candidate, element) used
A: T_0, E_2, E_3 B: T_1, H_5 C: H_4

the physics is in the difference: H_4 was linked to both tracks but used by neither; T_0 was linked to E_3 and kept it

Where to read more

- `Reco2Pico/docs/TrueJES.md`: the physics note
- `Reco2Pico/PicoProducer/README.md`: the table definitions
- `Reco2Pico/analysis/TrueJES/README.md`: the analysis